RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Thursday, 20 August 2026, 10:42 AM
State	Finished
Completed on	Thursday, 20 August 2026, 10:45 AM
Time taken	2 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Recursive function utilizing divide and conquer two-pointer strategy
int find_pair_sum(int arr[], int left, int right, int x, int *res1, int *res2) {
    // Base Case: If pointers cross each other, no valid pair exists
    if (left >= right) {
        return 0; // Return 0 indicating failure
    }

    int current_sum = arr[left] + arr[right];

    // Condition 1: Target sum found
    if (current_sum == x) {
        *res1 = arr[left];
        *res2 = arr[right];
        return 1; // Return 1 indicating success
    }

    // Condition 2: Sum is smaller than target, move left pointer rightward
    else if (current_sum < x) {
        return find_pair_sum(arr, left + 1, right, x, res1, res2);
    }

    // Condition 3: Sum is larger than target, move right pointer leftward
    else {
        return find_pair_sum(arr, left, right - 1, x, res1, res2);
    }
}

int main() {
    int n, x;

    // Read the size of the array
    if (scanf("%d", &n) != 1) return 0;

    // Dynamically allocate memory for the array
    int *arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) return 1;

    // Read n elements of the array
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            free(arr);
            return 0;
        }
    }

    // Read the target sum x
    if (scanf("%d", &x) != 1) {
```

```
51          free(arr);
52          return 0;
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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